

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2013

EC305.01 Microcontroller & Application

Date: 27.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Differentiate general microprocessors and microcontroller. [02]
(b) Write a program to generate a square wave of 25% duty cycle on pin 2.3. [05]
- Q - 2 (a) List advantages of C language over assembly language. Also with help of suitable example explain data types of C. [07]
(b) List SFRs available in 8051 microcontroller. Draw and explain format of TMOD register. [07]

OR

- Q - 2 (a) Write an C program to toggle all the bits of P0,P1 and P2 continuously with a 50 ms delay using timer 0 mode 1. XTAL = 12 MHz. [07]
(b) Write an C program to measure frequency of signal applied at P3.4. XTAL = 22 MHz. [07]
- Q - 3 (a) i. How can the baud rate of data transfer be doubled? Along with it, write instructions for same. [03]
ii. Differentiate serial communication and parallel communication. [04]
- (b) Two switches are connected to pins P3.2 and P3.3. When a switch is pressed, the corresponding line goes low. Write a program to
i. Light all LEDs connected to port 0, if the first switch is pressed.
ii. Light all LEDs connected to port 2, if the second switch is pressed. [07]

OR

- Q - 3 (a) i. Differentiate Level-triggered interrupt and edge trigger interrupt. [03]
ii. Explain IE register of 8051 microcontroller. [04]
- (b) Write an 8051 C program to transfer serially the message "The earth is one country and mankind its citizens" continuously at a 4800 baud rate. XTAL=11.0592MHz. [07]

SECTION – II

Q - 4 Interface 16×2 LCD with 8051. Also write assembly or C program to display "CSPIT" in first line and "Changa" in second line. [07]

Q - 5 (a) Explain with diagram of interfacing, ADC 0804 in self clocking mode interfaced with 8051. Also write a program to display of output on LED. [07]

(b) Give classification of memory available in real world. [07]

OR

Q - 5 (a) Interface DAC0808 with 8051 and connect a switch to pin P0.0. Write a program to do following, [07]

i. If SW = 0, the DAC output gives a staircase waveform.

ii. If SW = 1, the DAC output gives a triangular waveform.

(b) Interface 16KB program ROM and 16KB data RAM with 8051. [07]

Q - 6 Attempt any two. [14]

(a) Two switches (SW0 and SW1) are connected to pin P2.0 and P2.1 respectively. Write an assembly language program to monitor the status of SW and perform the following with diagram of interfacing,

i. If SW0 is pressed, the stepper motor moves clockwise

ii. If SW1 is pressed, the stepper motor moves anticlockwise.

iii. If both switches are pressed simultaneously, motor stops.

(b) A switch is connected to pin P2.7. with help of neat diagram, Write a program to monitor the status of SW and perform the following:

i. If SW = 0, the DC motor moves clockwise.

ii. If SW = 1, the DC motor counterclockwise.

(c) List features of DS12887 RTC. Interface DS12887 with 8051 and explain DS12887 address map and bit format of register A.
